

A136 · Performance of PTM Identification Strategies in Mass Spectrometry Proteomics Search

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While mass spectrometry proteomics can assign post-translational modifications (PTMs), considering large numbers of PTMs balloons the search space, limiting statistical power and assignment confidence. Existing methods to augment proteome databases with PTMs include searching for modifications on a set of amino acids (variable search), searching on previously characterized PTM sites, and searching for peptide matches with mass shifts corresponding to the mass of a PTM (open search). We developed several strategies utilizing characterized sites to determine additional likely modified sites. A motif-based approach was intended to limit the search space to the local contexts in which PTMs are known to occur. Other approaches limit possible sites to proteins or protein ontologies with known PTM sites. Each PTM strategy was benchmarked with a brain proteomics dataset and the MetaMorpheus search engine. 3 PTMs were searched for simultaneously, with 2 or 5 PTMs allowed per peptide. Variable search consistently produced the most PTM identifications, with motifs and open search generally performing better than other methods. However, this performance came at the cost of slightly higher peptide ambiguity and lower localization confidence. Additionally, these strategies also confidently identified many non-existent (i.e. “entrapment”) PTMs, suggesting that a portion of these identifications are false positives. Finally, to evaluate the impact of varying search space sizes, each strategy was tested on a ground truth set of known synthetic modified peptides. Overall, these studies suggest ensuring all possible PTM sites are in the database produces better results than shrinking the search space to exclude unlikely sites.

Performance of PTM Identification Strategies in Mass Spectrometry Proteomics Search

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